- **Optimizer**: The Adam optimizer adapts the learning rate and accelerates convergence.
- **Loss Function**: Mean Squared Error (MSE) is commonly used for regression tasks.
- **Metrics**: Mean Absolute Error (MAE) provides additional feedback on model performance.

Step 5: Training the Model

Train the model using the `fit` method. This method iterates over the dataset for a specified number of epochs.

```
history = model.fit(X_train, y_train, epochs=50, batch_size=32, validation_split=0.2)
```

- **Epochs**: Number of times the model sees the entire training dataset.
- **Batch Size**: Number of samples processed before updating the model parameters.
- **Validation Split**: Percentage of training data reserved for validation.

Step 6: Evaluating the Model

Evaluate the model on the test data to measure its performance.

```
eval_metrics = model.evaluate(X_test, y_test)
print("Loss:", eval_metrics[0])
print("Mean Absolute Error:", eval_metrics[1])
```

- **Loss**: Measures the error in predictions.
- **MAE**: Provides insight into the average magnitude of errors.

Enhancing the Neural Network

Adding Dropout Layers

Dropout layers help prevent overfitting by randomly disabling neurons during training.

```
from tensorflow.keras.layers import Dropout

model = Sequential([
    Dense(64, activation='relu', input_shape=(X_train.shape[1],)), Dropout(0.2),
```